# BT201 CURRENT SUBJECTIVE

**2 marks.**

1. **Cuvier's advantage On natural selection?**

Cuvier's anatomical and classificational work was impeccable.

- He identified four basic body plans for animals– vertebrates, molluscs, arthropods, and radiata (jellyfish & others).
- With these groups clearly distinguished, the chain of being was no longer part of biology.
- But links between distinct groups were still striking– Lamarck cited teeth in fetal baleen whales, for instance, which St. Hilaire took as an instance of 'unity of plan'. A broader sort of unity could restore the chain to some respectability…

2. **Protozoan characters?**
    - All are unicellular eukaryotes
    - Many species are both heterotrophic and autotrophic simultaneously or at different stages of the lifecycle
    - Protozoans are ecologically important primary producers, consumers and as vital links in the food chain
    - Humans are greatly effected by parasitic protozoans either directly or indirectly
    - Effects range from irritating - fatal
    - Malaria (*Plasmodium spp.*) worldwide epidemic
3. **Incomplete fossil record of soft bodies?**

The fossil record is relatively incomplete for several reasons:

1. Soft tissues are rarely preserved
2. Movement of the earth crust has obliterated or erased and or covered many fossils.
3. Fossilization takes place only in certain types of habitats and favorable environments
4. Paleontologists have not dug up every place

4. **What are the processes of the ecosystem influenced by landscape structure?**

Landscape structure influences processes such as the flow of energy, materials, and species between the ecosystems within a landscape.

**Examples**

Studies of the movements of small mammalsin a prairie landscape show that a smaller proportion of individuals moves in more fragmented landscape but that theindividuals that do move will move farther.

5. **Hardy weinberg principal?**

One of the most fundamental concepts in population genetics, the Hardy-Weinberg principle, states that "in a population mating at random in the absence of evolutionary forces, allele frequencies will remain constant."

**3marks**

1. **Proterozoic fossils?**

- Although scarce, a few animal fossils older than those of the Ediacaran fauna are known.
- A jellyfish-like impression is present in rocks 2000 m below the Ediacara Hills Pound Quartzite.
- Burrows in many areas presumably made by worms, occur in rocks at least 700 million years old.
- Wormlike and algae fossils come from 700 to 900 million-year-old rocks in China but the identity and age of these "fossils" has been questioned

2. **Advantage of being multicellular?**

For something on the order of 1.5 billion years

- all organisms were single-celled and life seems to have thrived .In fact, single-celled organisms are quite good at what they do but what they do is very limited .Single celled organisms can not grow very large, because as size increases proportionately less of a cell is exposed to the external environment in relation to its volume and the proportion of surface area decreases Transferring materials from the exterior to the interior becomes less efficient .

while

- multicelled organisms have cells specialized to perform specific functions such as respiration, food gathering, and reproduction
- multicelled organisms live longer, since cells can be replaced and more offspring can be produced
- Cells have increased functional efficiency when they are specialized into organs with specific capabilities

3. **How nutrient enrichment says the human is altering the aquatic and terrestrial ecosystem?**

Nutrient enrichment by humans is altering aquatic and terrestrial ecosystems.

- ***Nitrate*** concentration and export by the earth's major rivers correlate directly with human population density.
- Human disturbance also increases export of ***phosphorus*** from aquatic catchments.
- Nutrient enrichment appears to be reducing the diversity of plants and fungi in terrestrial ecosystems.

4. **Optimal foraging theory?**

Evolutionary ecologists predict that if organisms have limited access to energy, natural selection is likely to favor individuals that are more effective at acquiring energy and nutrients. Many animals select food in a way that appears to maximize the rate at which they capture energy.

Plants appear to allocate energy to roots versus shoots in a way that increases their rate of intake of the resources that limit their growth. Plants in environments with abundant nutrients but little light tend to invest more energy in the growth of stems and leaves and less in roots. In environments rich in light but poor in nutrients, plants tend to invest more energy in the growth of roots.

5. **How to documented genetic history in genome ?**

- Comparing nucleic acids or other molecules to infer relatedness is a valuable tool for tracing organisms' evolutionary history
- DNA that codes for rRNA changes relatively slowly and is useful for investigating branching points hundreds of millions of years ago
- mtDNA evolves rapidly and can be used to explore recent evolutionary events

6. **. EUsocialistic characteristics?**
    - The more complex level of social behavior, which is considered to be the pinnacle of social evolution, is called eusociality.
    - Eusociality is generally thought to include three major characteristics:

(1) individuals of more than one generation living together,
(2) cooperative care of young, and
(3) division of individuals into sterile, or nonreproductive, and reproductive castes.

**5marks**

1. **Lyell's Mistake on natural selection?**

As a part of his general resistance to direction/ substantial change in geology, Lyell maintained there was no real direction of change in the fossil record either. This was greeted with pretty general astonishment, but Lyell claimed the apparent changes were just due to our having bad (misleading) samples…more mammals would soon appear in older formations, etc., as we extend our fossil collections. Within ten years (1840) it was clear that this was completely untenable.

1. **Food web?**

**A food web summarizes the feeding relations in a community.**

- The earliest work on food webs concentrated on simplified communities in areas such as the Arctic islands. However, researchers such as Charles Elton (1927) soon found that even these so-called simple communities included very complex feeding relations.
- The level of food web complexity increased substantially, however, as researchers began to study complex communities. Studies of the food webs of tropical freshwater fish communities revealed highly complex networks of trophic interaction that persisted even in the face of various simplifications.
- A focus on strong interactions can simplify food web structure and identify those interactions responsible for most of the energy flow in communities.

**Food web and structure of communities**

- **The feeding activities of a few keystone species may control the structure of communities.**

Robert Paine (1966) proposed that the feeding activities of a few species have inordinate influences on community structure. He predicted that some predators may increase species diversity by reducing the probability of competitive exclusion.

Manipulative studies of predaceous species have identified many keystone species, including starfish and snails in the marine intertidal zone and fish in rivers. On land, birds exert substantial influences on communities of their arthropod prey.

2. **Endosymbiotic theory and origin of eukaryotic evolution?**

Formed from several prokaryotic cells . In a symbiotic relationship Symbiosis involving a prolonged association of two or more dissimilar organisms, is quite common today. In many cases both symbionts benefit from the association as occurs in lichens, once thought to be plants but actually symbiotic fungi and algae

**Evidence for Endosymbiosis**

Supporting evidence for endosymbiosis comes from studies of living eukaryotic cells containing internal structures called organelles, such as mitochondria and plastids which contain their own genetic material. In addition, prokaryotic cells synthesize proteins as a single system, whereas eukaryotic cells are a combination of protein-synthesizing systems.

**Organelles Capable of Protein Synthesis**

That is, some of the organelles within eukaryotic cells are capable of protein synthesis. These organelles with their own genetic material and protein-synthesizing capabilities are thought to have been free-living bacteria that entered into a symbiotic relationship, eventually giving rise to eukaryotic cells

3. **Explain with example how exotic predators disturb the structure of food web. ?**

**Exotic predators can collapse and simplify the structure of food webs**. Introduced fishes have devastated the native fishes of Lake Atitlan and Gatun Lake in Central America.

**Examples**

The influence of the Nile perch on the fish community of Lake Victoria is enmeshed with massive changes in the lake's ecosystem. Introduction of the Nile perch is rapidly reducing the species-rich fish fauna of Lake Victoria to a community dominated by a handful of species.

**Humans**

Humans have acted as keystone species in communities. People have long manipulated food webs both as a consequence of their own feeding activities and by introducing or deleting species from existing food webs. In addition, many of these manipulations have been focused on keystone species. Hunters in tropical rain forests have been responsible for removing keystone animal species from large areas of the rain forests of Central and South America. Chinese farmers have used ants as keystone predators to control pests in citrus orchards for over 1,700 years.

**10 marks:**

**Define Codon ,nonsense codon and describe the relationship beteen genotype and phenotype?**

CODON-A codon is a sequence of three DNA or RNA nucleotides that corresponds with a specific amino acid or stop signal during protein synthesis. DNA and RNA molecules are written in a language of four nucleotides; meanwhile, the language of proteins includes 20 amino acids. Codons provide the key that allows these two languages to be translated into each other. Each codon corresponds to a single amino acid (or stop signal), and the full set of codons is called the genetic code.

NONSENSE CODON-There are 3 codons out of 64 in genetic code which do not encode for any Amino Acid.  These are called termination codons or stop codons or nonsense codons. The stop codons are UAA, UAG, and UGA. They encode no amino acid. The ribosome pauses and falls off the mRNA. They signal release of the nascent polypeptide from the ribosome because no cognate tRNA has anticodons complementary to these stop signals, allowing a release factor to bind to the ribosome instead

Difference Between Genotype and Phenotypes

The genotype is the set of genes in our DNA which is responsible for a particular trait. Genotype is the genetic makeup of an organism and it results in some of the physical characteristics of that organism.The entire genetic information about an organism is contained in a genotype.. Genotype is determined by the makeup of something called "alleles

The phenotype is the physical expression, or characteristics, of that trait. For example, two organisms that have even the minutest difference in their genes are said to have different genotypes. These two mice may have different genotypes and both can still be white if they share the particular phenotype for white fur.

The phenotype is the physical appearance of an organism, while the Genotype is the genetic composition of an organism. Phenotype are observable and are the expression of the genes of an individual. So even the organism with the same species may differ, with a minute difference in their genotype. This is the main difference between the two.

We can notice one's hair colour, eye colour, height, weight, skin colour, etc. but cannot look at genes responsible for these characters, so the observable physical look is the phenotype while the unnoticed genes responsible for such characters present in the DNA of cell of the individual is genotype.

(From ppts)Populations include genetic and phenotypic variation among individuals, which are significant aspects of population structure. The first biologists to conduct thorough studies of phenotypic and genotypic variation and to incorporate experiments in their studies, focused on plants. By using plants cloned from the same parent, Bonnier was able to control for the effects of genotype on plant form and observe the effects of climate. .

## Molecular Basis for
Relationship between Genotype and Phenotype

